

SIMATS ENGINEERING

ASSESSMENT TOOL 2

Case study

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Code of Conduct:

I Saree BL 192524023 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

B. S.

QUESTIONS: 4

1. A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss**, **180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2. A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS**, **DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and reliability.

3. A cloud service provider serves millions of users worldwide. During promotional events, users experience significant delays in accessing services. Network analysis reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants to improve its DNS infrastructure to handle global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning

项目	名称	规格	数量	单位	备注
材料费	水泥	42.5	100	m³	
	砂	中砂	200	m³	
	石子	20mm	150	m³	
	钢筋	HPB300	10	t	
	木材	杉木	5	m³	
人工费	木工	10	工日		
	泥工	10	工日		
	钢筋工	10	工日		
	架子工	10	工日		
	其他	10	工日		
机械费	搅拌机	100L	1	台班	
	电焊机	200A	1	台班	
	切割机	400W	1	台班	
	打桩机	1000N	1	台班	
	其他	1000W	1	台班	

(i) Nagle's Algorithm.

- * Combines small TCP packets.
- * Reduces network overhead.
- * Can delay small trading messages.
- * This is undesirable for low latency trading.

(ii) SYN Queue.

- * Stores incoming connection requests.
- * High traffic can fill the SYN queue.
- * Cause connection delays.
- * May result in SYN retransmission.

2. Effect on Transaction Processing.

- * Flow control can slow data transmission.
- * Nagle can delay small order messages.
- * SYN queue limits delay new connections.
- * Retransmissions increase latency.
- * Buffering delays order acknowledgments.
- * These factors can increase latency from 1ms to 40ms.

3. Recommended TCP Changes.

- * Enable TCP_NODELAY where appropriate.
- * Increase SYN queue backlog capacity.
- * Optimize TCP send and receive buffers.
- * Use persistent TCP connections.
- * Reduce unnecessary buffering.
- * Monitor TCP re-transmission.
- * Reduce network congestion.
- * Prioritize trading traffic using QoS.

③ TTL, Anycast and Caching

TTL * Controls how long DNS records are cached

* Higher TTL reduces DNS queries.

* Lower TTL allows faster record updates.

Anycast:

* Uses the same IP from multiple locations

* Direct users to a suitable DNS server.

* Reduces lookup latency.

* Improves fault tolerance.

Caching:

* Stores DNS responses temporarily

* Reduces repeated DNS lookups.

* Reduces server workload.

* Improves response speed.

Stock trading systems

④ Three TCP Factors

(1) Flow Control

* Controls the amount of data sent.

* Prevents receiver overload.

* Small receive window can reduce performance.

* Can cause buffering and delays

Distributed DNS

① Impact of Centralized DNS

- * All queries go to one DNS server
- * Creates a single point of failure
- * Causes server overload.
- * Increases DNS lookup time
- * Reduces application performance.
- * Poor scalability during traffic spikes
- ↓ Failure can affect millions of users

② Distributed DNS Solution

- * Deploy DNS server in multiple regions
- * Use Anycast routing
- * Use primary and secondary DNS servers
- * Enables DNS caching.
- * Monitor DNS server health.
- * Redirect traffic when a server fails.
- * Use geographically closer DNS servers
- * Distribute queries across multiple servers

(ii) Proposed DNS Architecture

- * Use Anycast DNS
- * Deploy DNS servers in different regions.
- * Use DNS caching.
- * Implement DNS pre-fetching.
- * Optimize TTL values.
- * Use multiple authoritative DNS servers.
- * Provide automatic failover.
- * Target DNS latency below 50 ms

(iii) Advantages

- * Lower DNS latency.
- * Better global performance.
- * High availability.
- * Better scalability.
- * Automatic failure handling.

(iv) Limitations

- * High Infrastructure cost.
- * More complex management.
- * Anycast requires proper routing.
- * Incorrect TTL can cause stale records.
- * Pre-fetching creates additional DNS traffic.

2. Improvements

- * Use UDP-based real-time transmission.
- * Apply QoS for video/audio traffic.
- * Use adaptive bitrate.
- * Reduce video quality during congestion.
- * Use jitter buffers.
- * Use packet loss concealment.
- * Apply forward error correction where suitable.

2. DNS Infrastructure

1. Reasons for High DNS Resolution Time

- * DNS lookup takes about 800ms.
- * DNS server may be geographically far from user.
- * DNS server may be overloaded.
- * Poor DNS caching increases lookup time.
- * Low TTL causes frequent DNS queries.
- * Network congestion can delay DNS response.
- * Multiple DNS resolution steps increase latency.

① Video Conferencing

i) TCP / UDP or Application Layer

* The main issue is network congestion and packet loss.

* Packet loss is $\approx 1\%$ and jitter is 180ms

* UDP is generally suitable for real time video

* TCP retransmission can increase delay.

* Application buffering also affects video quality.

* Therefore both transport and application factors contribute

(ii) Requirements of Real Time Multimedia

* Requires low latency.

* Requires low jitter.

* Requires stable bandwidth.

* Small packet losses can be tolerated.

* UDP avoids re-transmission delays.

* Audio/Video should arrive quickly rather than perfectly.

Q.2 * Adapting buffering improve playback